

	site 1	site 2	site 3	site 4
χ^1	X	I	I	I
χ^2	Y	I	I	I
χ^3	Z	X	I	I
χ^4	Z	Y	I	I
χ^5	Z	Z	X	I
χ^6	Z	Z	Y	I
χ^7	Z	Z	Z	X
χ^8	Z	Z	Z	Y

Z Jordan-Wigner string

X, Y the active site

I identity

$$\{\chi_i, \chi_j\} = \delta_{ij} \cdot I$$

verified to 2.2e-16

the 1/√2 normalisation gives

$$\chi_i^2 = \frac{1}{2} \cdot I \quad (\text{to } 1.1\text{e-}16)$$

N = 8 Majoranas on 4 qubits — Hilbert dimension $2^4 = 16$